Before and After Slides

Six worked redesigns. All data are fictional.

Neudata

Insight. Impact. Innovation.

BEFORE

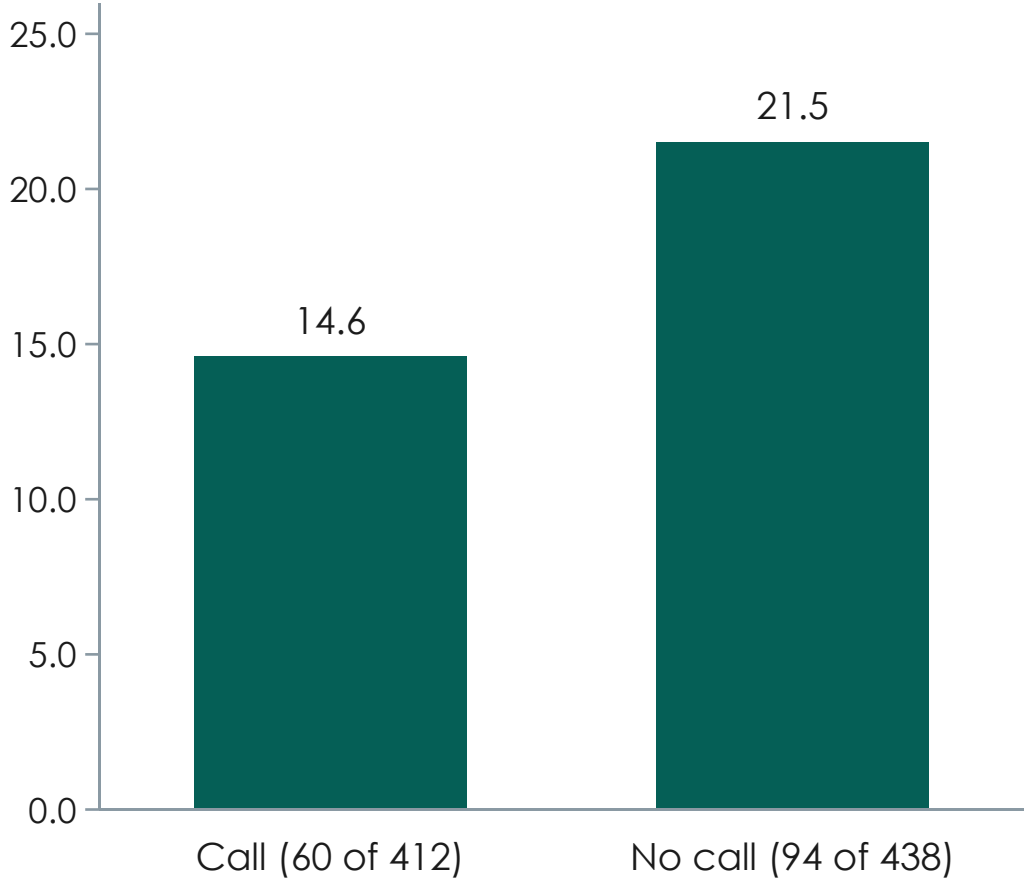
Results

- Total sample size was 850 patients discharged from medical wards at Hospitals A and B during the study period
- 412 patients received a nurse telephone call and 438 patients did not receive a nurse telephone call
- In the call group 60 patients (14.6%) had an unplanned readmission within 30 days
- In the no-call group 94 patients (21.5%) had an unplanned readmission within 30 days
- The risk difference was 6.9 percentage points (95% CI 1.8 to 12.0)
- Patients at high discharge risk made up 29% of the call group and 45% of the no-call group
- After stratification by discharge risk the difference was 4.3 points (9.4 lower to 0.8 higher)
- This suggests that the intervention may have had a beneficial impact on readmission outcomes

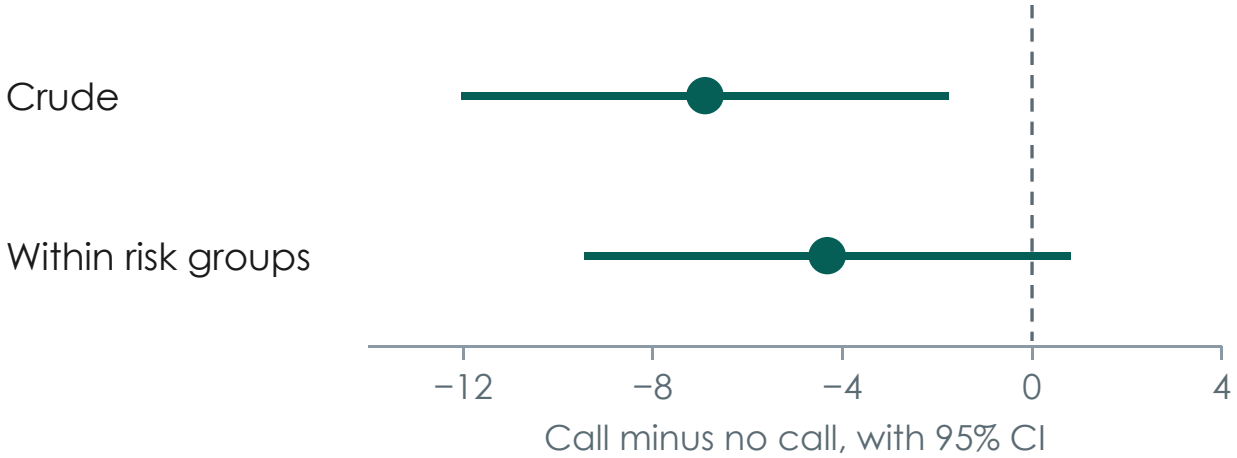
Fictional teaching data, not a real study.

AFTER

Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk



Difference in readmissions per 100 patients



High-risk patients were harder to reach: 29% of those called, 45% of those not called.

Calls were not randomised.

Fictional teaching data, not a real study.

BEFORE

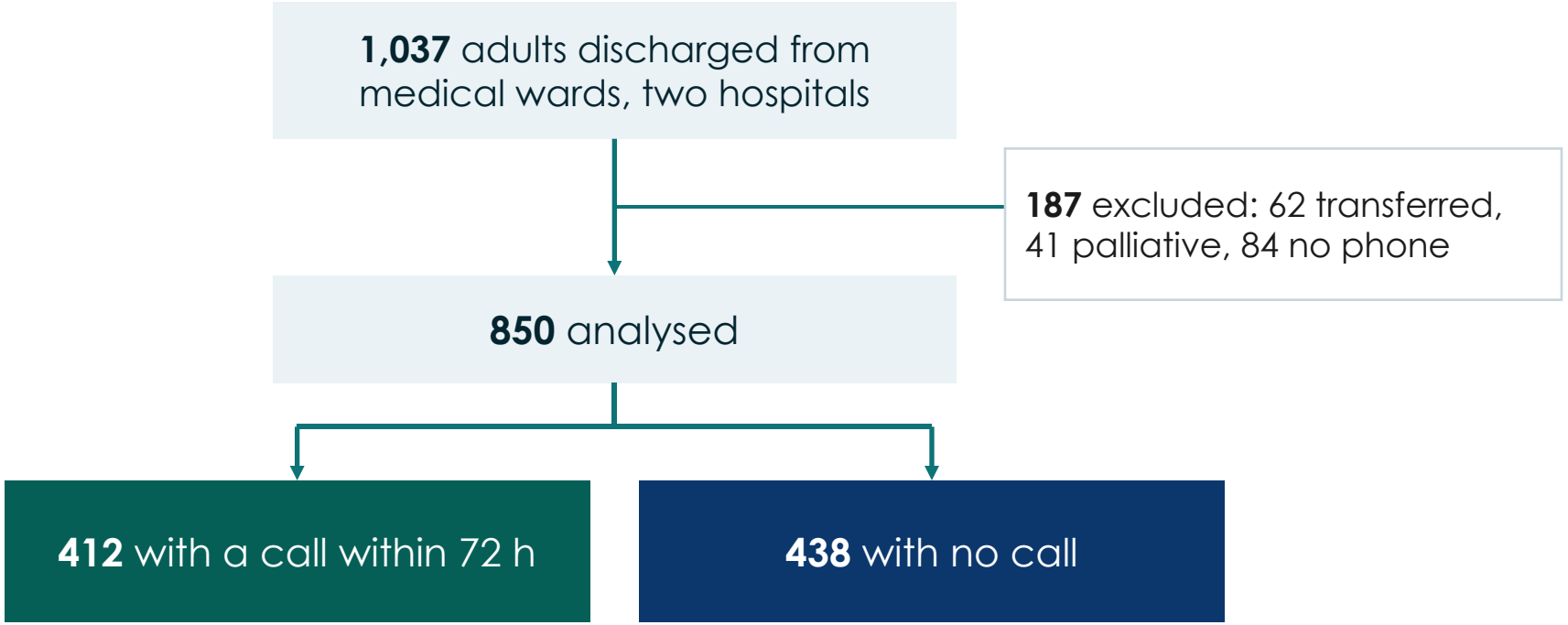
Methods

- Retrospective observational cohort study
- Two hospitals, medical wards, one calendar year
- Inclusion criteria: adults discharged home
- Exclusion criteria: transfer (n=62), palliative pathway (n=41), no working phone number (n=84)
- Exposure: nurse telephone call within 72 hours of discharge
- Outcome: unplanned readmission within 30 days
- Statistical analysis: risk differences, 95% Wald confidence intervals, Mantel-Haenszel stratification by discharge risk (high or lower)

Fictional teaching data, not a real study.

AFTER

We compared 850 patients who did or did not get a call within 72 hours



Why it matters

Nurses called when free and when a patient could be reached, so groups may differ.

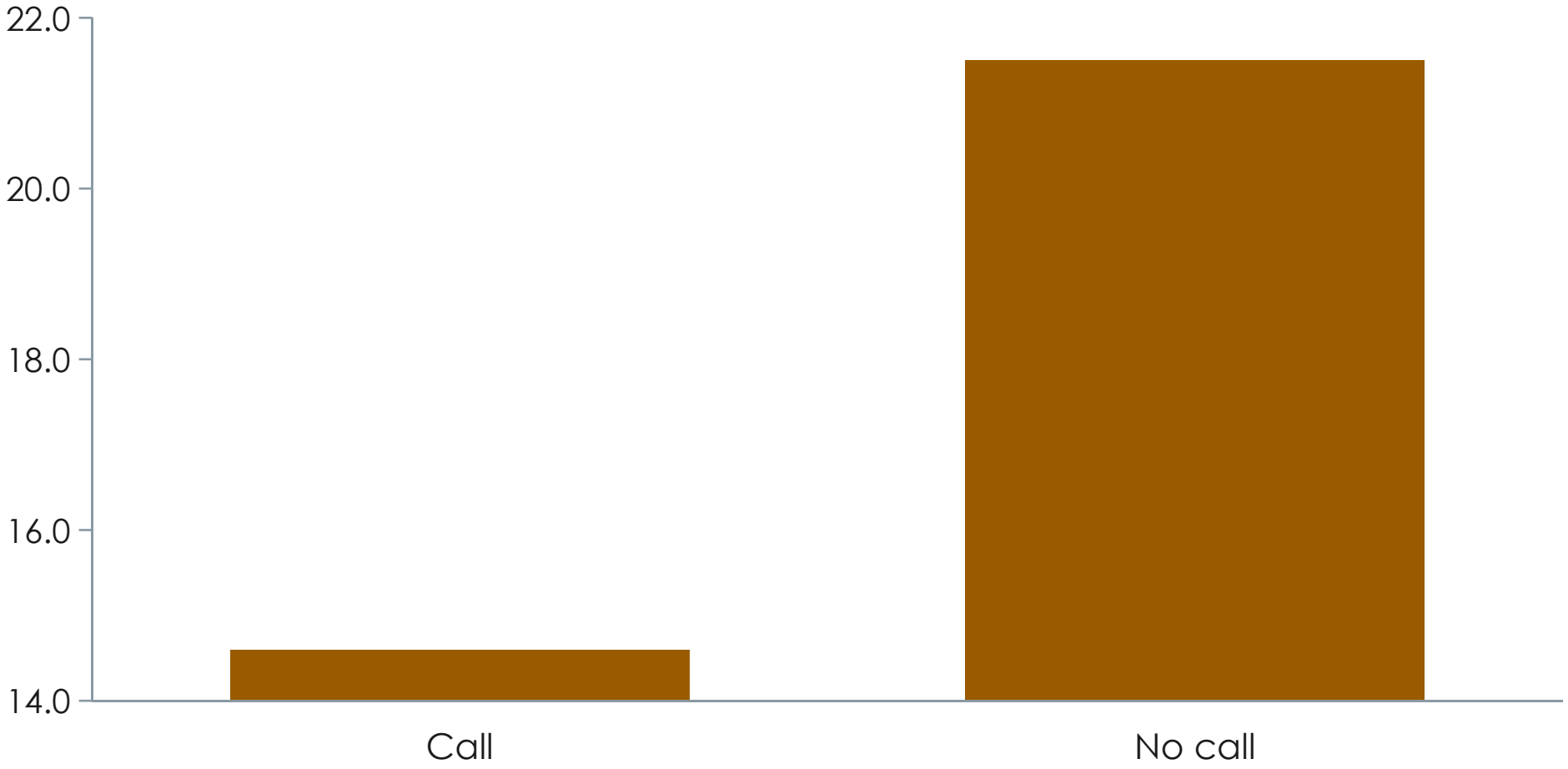
Outcome: unplanned readmission within 30 days.

Analysis: risk difference, overall and within discharge-risk groups.

Fictional teaching data, not a real study.

BEFORE

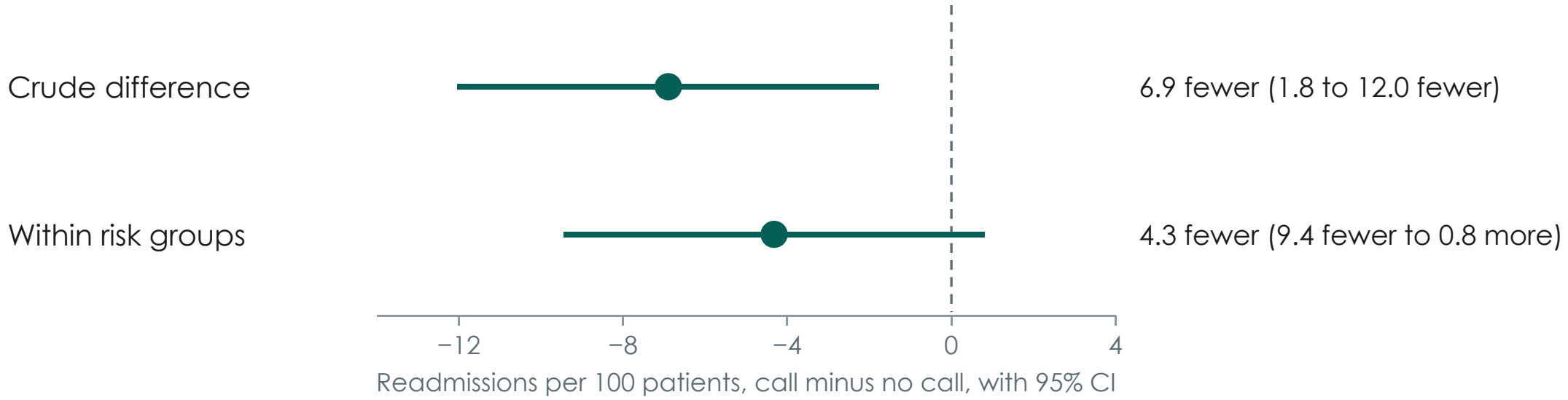
Calls produce a massive drop in readmissions!



Fictional teaching data, not a real study.

AFTER

Readmissions were 6.9 per 100 lower with a call, but the data allow a wide range



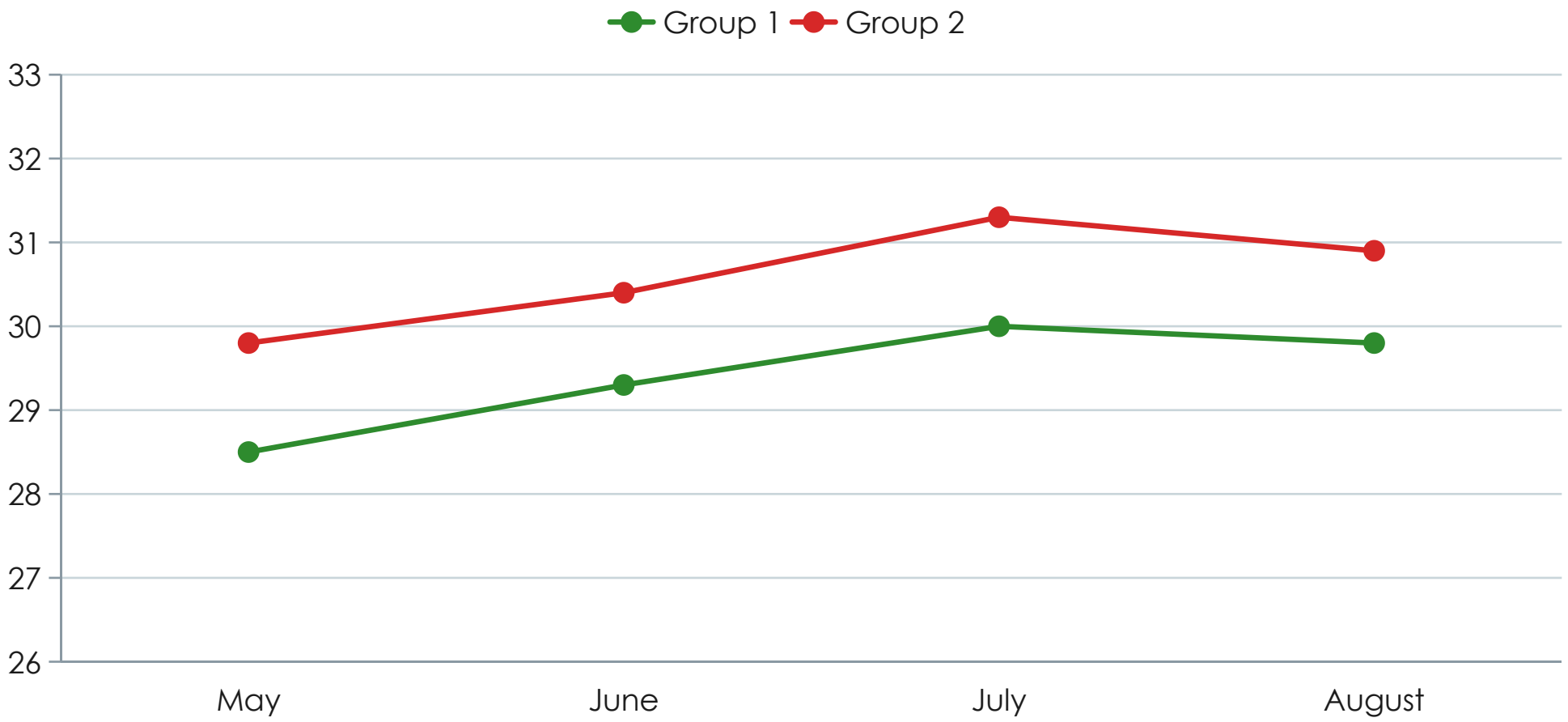
Risk of readmission: 60 of 412 (14.6%) with a call, 94 of 438 (21.5%) without.

The dot uses position, so the axis can focus on the range of the differences.

Fictional teaching data, not a real study.

BEFORE

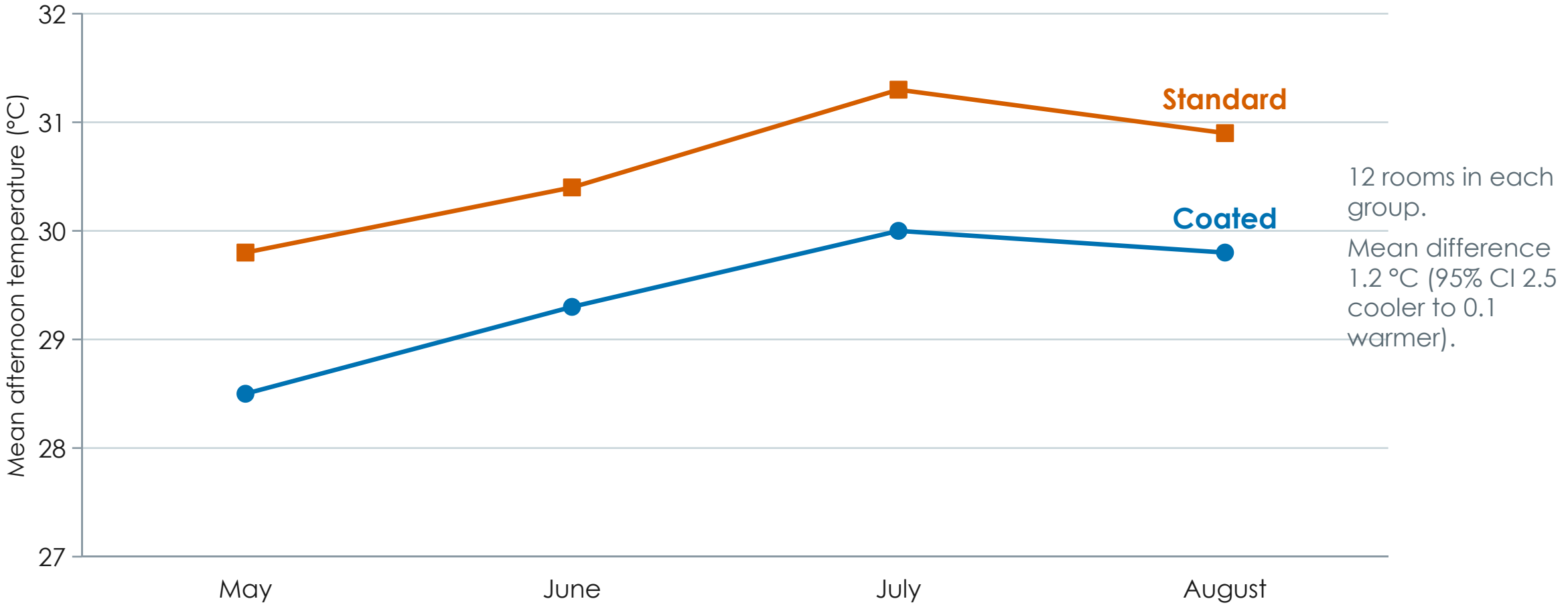
Temperature comparison by month



Fictional teaching data, not a real study.

AFTER

Coated classrooms averaged 1.2 °C cooler, but the interval includes no difference



12 rooms in each group.

Mean difference 1.2 °C (95% CI 2.5 cooler to 0.1 warmer).

Fictional teaching data, not a real study.

BEFORE

Conclusions

- Nurse telephone follow-up is a promising intervention
- Further research is needed
- Limitations of this study include its observational design
- The team would like to acknowledge everybody who helped

Thank you! Any questions?

Fictional teaching data, not a real study.

AFTER

Recommendation: test calls on two wards for six months before funding full roll-out

What we know

4.3 fewer readmissions per 100 within risk groups.

The data fit 9.4 fewer to 0.8 more.

What it costs

About 137 nurse hours for 412 calls at 20 minutes each.

What we do not know

Whether calls caused the difference, because reachable patients were healthier.

Decision needed today

Approve a staggered six-month trial on two wards, so that allocation no longer depends on who can be reached. Decide on full roll-out from its results.

Fictional teaching data, not a real study.

BEFORE

Subgroup analysis

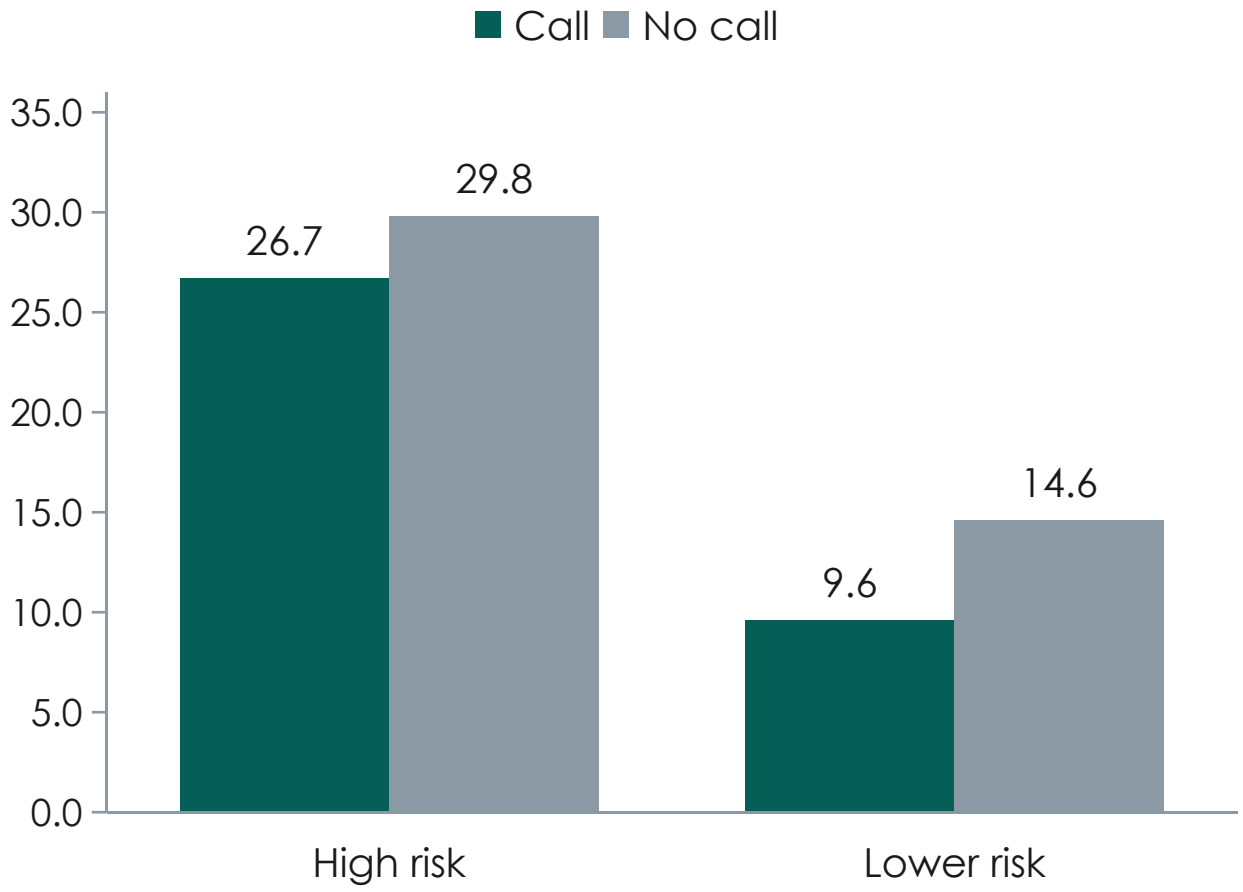
Stratum	Call events	Call n	Call %	No-call events	No-call n	No-call %	p
High	32	120	26.7	59	198	29.8	NS
Lower	28	292	9.6	35	240	14.6	NS

- Green = good, red = bad
- No significant difference in either stratum, so calls had no effect in subgroups
- However the overall analysis was significant
- See table for further details

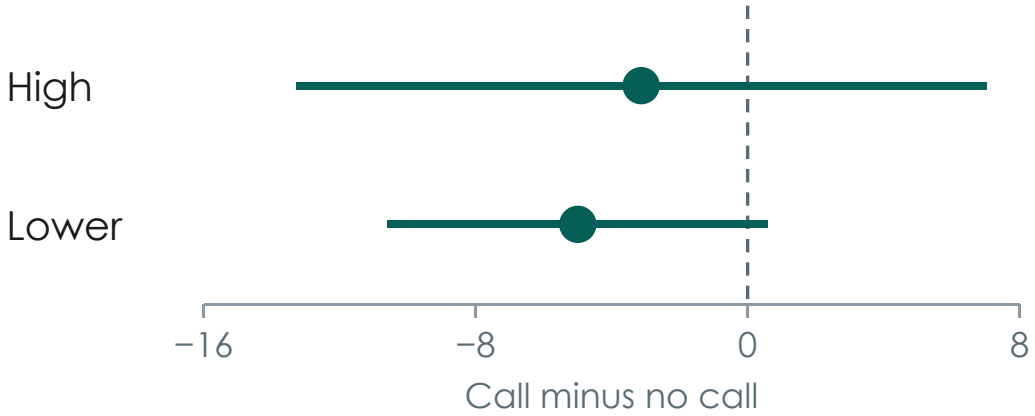
Fictional teaching data, not a real study.

AFTER

In both risk groups fewer patients who got a call were readmitted, and the intervals are wide



Difference per 100 (95% CI)



High risk: 32 of 120 v 59 of 198.

Lower risk: 28 of 292 v 35 of 240.

Too few patients to separate these groups.

Fictional teaching data, not a real study.



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